

MEWA — MULTI-MODAL EARLY WARNING ADVISOR

An early warning advisor for extreme weather and drought, field-tested with Bangladesh's farmers and *extension network*.

MEWA combines meteorological forecasts, satellite observation and a Bengali-language conversational interface. It locates a farmer's field, forecasts its weather, assesses drought risk, recommends crop actions and sends early warning alerts, all from one set of coordinates. After winning the GenAI for Good Challenge, MEWA went into the field: a pilot in Sapahar upazila, Naogaon district, with 100 farmers and agriculture extension officers, run with BRAC and the Department of Agricultural Extension.

100
PILOT PARTICIPANTS IN SAPAHAR

4
PILOT CROPS: RICE, MANGO, TURMERIC, EGGPLANT

11
WEEKS, SETUP TO EVALUATION

v1.0
RELEASE AFTER THE PILOT

01 — EXECUTIVE SUMMARY

Weather and drought advisories from national agencies rarely reach farmers in *remote districts* of Bangladesh.

MEWA, the Multi-modal Early Warning Advisor, is a conversational early warning system for extreme weather and drought risk, built by Polisense AI as an application on the International Telecommunication Union's open source GENIE.AI platform. For one set of coordinates, it locates the field, forecasts its weather, assesses drought risk, and turns both into early warnings and practical crop recommendations in Bengali, sent before conditions become damaging rather than after. In 2026, MEWA was named winner of the Climate category of the GenAI for Good Challenge, presented at the AI for Good Global Summit by the International Telecommunication Union and IEEE Humanitarian Technologies. After the award, MEWA was taken into the field: a pilot in Sapahar upazila, Naogaon district, with 100 participants and supported by BRAC's Climate Change Program, is described in section 04.

THE GAP MEWA ADDRESSES

Bangladesh's national agencies already produce the underlying advisories. The Bangladesh Meteorological Department issues forecasts and warnings; the Bangladesh Agro-Meteorological Information Service, known as BAMIS, publishes crop-specific calendars and bulletins. Both are technically sound and regularly updated. Both are also distributed as formatted documents and web pages built for institutional readers, not for a Sub-Assistant Agriculture Officer standing in a field with a mobile phone.

Today, extension officers close this gap themselves. Each covers on the order of nine hundred farmers, and the BAMIS crop calendars they rely on were developed under a World Bank project that no longer maintains them, so officers already work with material that is incomplete and unevenly updated. MEWA's chat interface and web dashboard serve extension officers, who relay guidance onward, the same pattern that BRAC's own farmer advisory service depends on. In the Sapahar pilot the mobile application also went directly into farmers' hands, so that advice could reach a field without waiting for a visit. Delivery by text message and voice, the channels most rural households already use, remains part of the work ahead.

> MEWA's original reference crop was potato, with a documented fifteen-week district calendar. For the Bangladesh pilot, crop profiles, seasonal calendars and risk thresholds were built for rice, mango, turmeric and eggplant, the crops Sapahar's farmers grow, and the approach extends to further crops as each is added.



FIG. 01 · MEWA's menu: crop advisory, weather alerts, irrigation guidance, subsidy information, open chat, and a handsfree voice mode.

IN THIS DOCUMENT

02
How MEWA responds to one location, step by step.

03
Recognition: the GenAI for Good Challenge win, and an early accuracy evaluation.

04 – 05
The Bangladesh pilot, then next steps: scaling, BRAC's role, and what remains open.

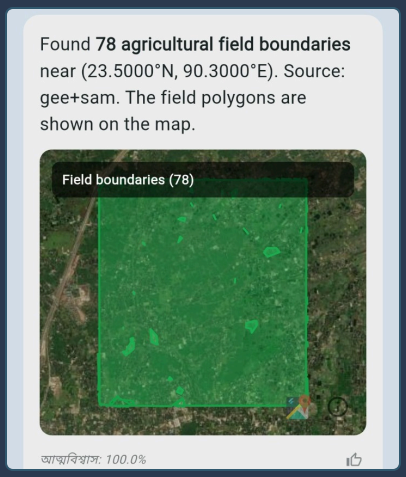
The five capabilities behind this menu, each grounded in a specific data source and tied to one location, are mapped step by step on the next page.

02 — HOW MEWA RESPONDS

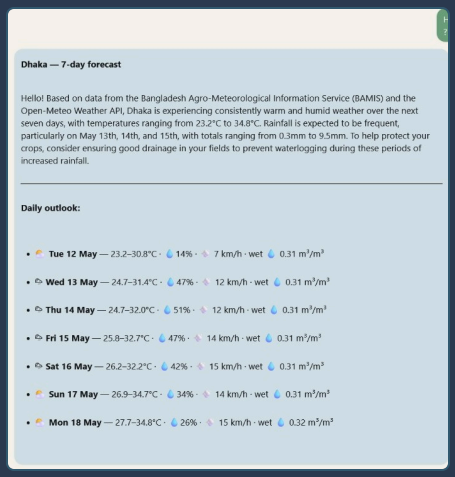
One set of coordinates becomes five capabilities: locate the field, forecast its weather, assess drought risk, recommend crop action, and send *proactive alerts*.

01 **Locate the field** → 02 **Forecast the weather** → 03 **Assess drought risk** → 04 **Recommend crop action** → 05 **Send proactive alerts**

Every step runs against the same location. MEWA finds the farmer's field from its coordinates, pulls the forecast and drought indicators for that exact spot, turns both into a crop recommendation, and keeps watching afterward so that a change in conditions triggers an alert without a new question being asked. The five responses below are real, drawn from the application.



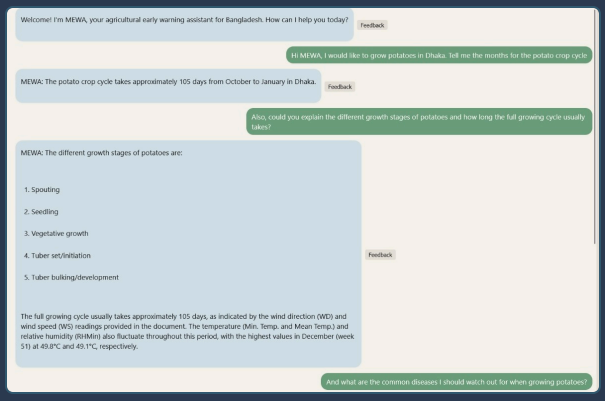
01 · LOCATE · 78 field boundaries returned for one coordinate pair.



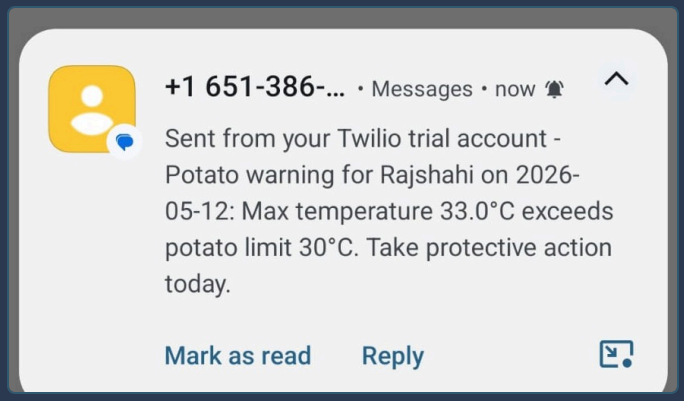
02 · FORECAST · a seven-day outlook for that same field.



03 · ASSESS · a drought outlook built from four indicators.



04 · RECOMMEND · a full potato crop cycle, stage by stage, for the same district.



05 · ALERT · a text message sent when a threshold is crossed, unprompted.

None of these five responses is generic. Each is computed for the coordinates the officer or farmer supplied, which is why the same conversation can move from a satellite map, to a weather table, to a drought score, to a recommendation, without the person ever repeating their location.

03 — RECOGNITION

In 2026, MEWA was named winner of the *Climate category* of the GenAI for Good Challenge.

The GenAI for Good Challenge is presented at the AI for Good Global Summit by the International Telecommunication Union and IEEE Humanitarian Technologies. Three category winners were named in 2026: Polisense AI, for MEWA, in the Climate category; Innov8AI in the Health category; and AgriConnect in the Agriculture category. The Food and Agriculture Organization, the World Health Organization and the World Meteorological Organization served as domain partners on the evaluation.

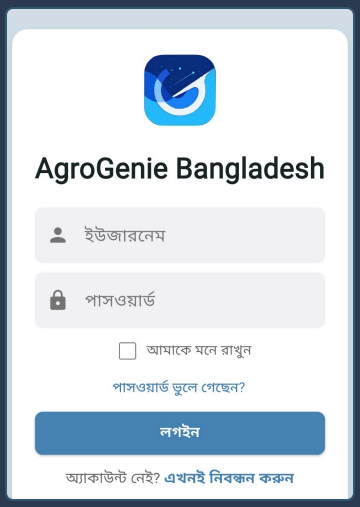


FIG. 03 — The Polisense AI team receiving the Climate category award at the AI for Good Global Summit: Bruno Galdos, José Pastor, Abhirup Das and Khuzaima Aziz.

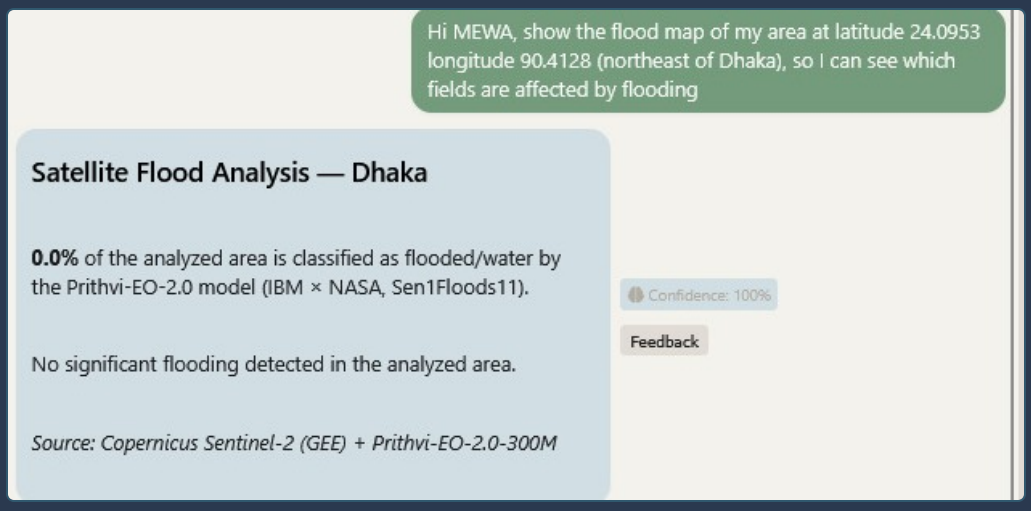
An early evaluation against twenty ground truth questions drawn from BAMIS bulletins scored sixty percent fully correct by an independent language-model judge, with a BERTScore of 0.84. This is a first read on a small test set, not a mature benchmark, and further evaluation is part of the next phase of work.

TEAM, PHASE 2

Bruno Galdos, technical lead. José Pastor, product and stakeholder coordination. Abhirup Das, mobile application and language services. Khuzaima Aziz, geospatial and drought analysis.



The mobile application, in Bengali throughout.



A flood analysis for a coordinate near Dhaka, using IBM and NASA's Prithvi-EO-2.0 model over Sentinel-2 imagery: a sixth capability alongside the five on the previous page.

04 — THE BANGLADESH PILOT

After the award, MEWA went into the field: a pilot with 100 participants in *Sapahar, Naogaon*.

From 15 August to 30 October 2026, Polisense AI is running a structured pilot of MEWA in Sapahar upazila, Naogaon district, with BRAC and the Department of Agricultural Extension supporting implementation and data collection. One hundred participants, farmers, extension officers or both, use the Bengali mobile application on their own phones, on four crops: rice, mango, turmeric and eggplant. The pilot set out to answer a practical question: can MEWA deliver weather, drought and crop-risk advice that is understandable, timely and useful under real local conditions?



FIG. 05 · A farmer in Sapahar opens MEWA Bangladesh in her vegetable plot.

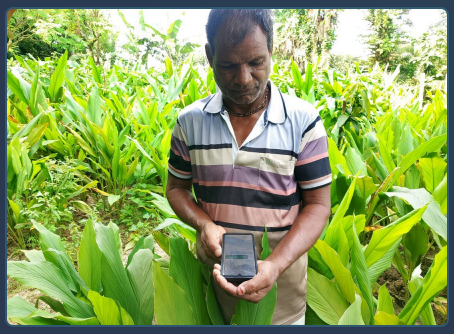


FIG. 06 · Reading an advisory on the phone among turmeric.



FIG. 07 · Checking a message at the edge of a rice field.

FIVE PHASES, 15 AUGUST TO 30 OCTOBER

15–31 AUG
Setup
Sapahar configured; weather, drought and crop-risk pipelines validated; Bengali interface and notifications tested.

1–15 SEP
Internal pilot
End-to-end testing with extension officers; alerts checked against BAMIS and BMD observations.

16–30 SEP
Field pilot
MEWA deployed in Sapahar with farmers and extension officers under real growing conditions.

1–15 OCT
Iteration
Bengali responses, crop thresholds, security and notification reliability improved from pilot feedback.

16–30 OCT
Evaluation
Alert accuracy, latency, delivery rate, engagement and satisfaction analysed; case study prepared.

Sapahar was chosen for its well-documented 2016 drought, which gives the drought-monitoring component a real event to calibrate against. The pilot concludes with a milestone on 1 November: MEWA v1.0, a deployment package, and an expansion roadmap for Naogaon District and the rest of Bangladesh.

WHAT THE PILOT MEASURED

Reach and use
Participants registered and receiving at least one advisory, active and repeat users, queries submitted, alerts received and opened, by weather, drought or crop risk.

Quality and delivery
Advisory delivery rate, response time, and extension-officer review of a sample of alerts against BAMIS, BMD and field observation: correct, partly correct or incorrect.

Usefulness and uptake
Participant ratings from 1 to 5 for clarity, relevance, actionability and timeliness; whether the recommendation was followed; satisfaction and willingness to keep using MEWA.

Data is drawn from system logs, a short endline survey, an alert review form completed by extension officers, and a field record of the advisory received and the action taken. Because the field test lasts two weeks, with 100 participants and no control group, it can establish reach, actual use, delivery reliability, perceived usefulness, recommendation uptake, user acceptance and preliminary alert agreement. It does not claim measured changes in yield, income, losses or water use.

05 — NEXT STEPS: SCALING IN BANGLADESH

Next, scaling MEWA across Naogaon and Bangladesh with *BRAC*.

BRAC's Climate Change Program already delivers weekly weather and agricultural advisories to more than twenty thousand farmers by recorded voice call, a format chosen because many farmers use basic phones rather than smartphones, together with WhatsApp groups run by lead farmers who relay information onward. Through stakeholder sessions organized with the International Telecommunication Union and IEEE, BRAC confirmed that the extension-officer relay model is a realistic distribution channel for MEWA, and helped identify the priority crops for the pilot. Boro rice, given its production value and exposure to flash floods, was the first priority, with Aman rice, wheat, potato and mango also considered; the pilot itself ran on rice, mango, turmeric and eggplant.

The Sapahar pilot gives the scaling work a tested starting point: validated crop profiles and thresholds, a Bengali interface used on farmers' own phones, and an evaluation report to guide the next districts. One item remains open: a formal data-sharing agreement with the Bangladesh Meteorological Department and BAMIS, whose crop calendars have no public interface and have been maintained informally since the World Bank project that built them ended.

HOW IT WORKS, BRIEFLY

MEWA extends the ITU's open source GENIE.AI platform. Risk classification is deterministic rather than left to a language model: fixed thresholds on soil moisture, evapotranspiration, rainfall runoff and vegetation greenness produce a five-tier risk level, and the language model's role is to explain that result in plain Bengali, not to compute it. Weather data comes from Open-Meteo, cross-checked against Bangladesh Meteorological Department and BAMIS figures; satellite data comes from Google Earth Engine and NASA's SMAP and MODIS instruments.

CLOSING

MEWA has now been used by farmers and extension officers in an actual growing season, not only in a demonstration. The next test is whether it holds at district scale, where an advisory still has to reach the right person in time to change a decision.

THE FIVE-TIER RISK SCALE BEHIND EVERY ALERT

1 · NORMAL	2 · ADVISORY	3 · WARNING	4 · SEVERE	5 · EMERGENCY

Warning level and above are the ones dispatched as proactive alerts, the same mechanism shown on page 3.

DATA SOURCES AND DELIVERY, AT A GLANCE

Meteorological

Open-Meteo for forecasts, cross-checked against the Bangladesh Meteorological Department and BAMIS.

Satellite

Google Earth Engine, NASA SMAP soil moisture and MODIS vegetation greenness.

Delivery today

Mobile application with push and SMS notifications, and a web dashboard, in Bengali, used by farmers and extension officers.

SUPPORTED BY: ITU · IEEE Humanitarian Technologies · brac · GenAI for Good CHALLENGE

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ALIGNED WITH SDGS: 2 ZERO HUNGER · 13 CLIMATE ACTION